CS454 Midterm 2 Part B

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***Abstract*—The exam is open book. No computers are allowed.**

I. Question

Propose a tree induction method where each split in each node is in the form of $a \leq x_i \leq b$.

II. Question

Consider the alternative error function in linear perceptron for regression problem:

$$E = \frac{1}{2}(r - y)^2 + \alpha \sum_i w_i^2 \tag{1}$$

where N represents the number of instances, r represents the actual output, y represents the calculated output, v_i are the weights between the input units and the output unit. Derive the gradient descent update rules for the weights.

III. Question

If each base-learner is independently and identically distributed and correct with probability $p >$ 1/2, what is the probability that a majority vote over L classifiers gives the correct answer? Calculate the probability for L = 5 and p = 3 / 4.

IV. Question

Consider the three learners d_1, d_2, and d_3; and the posterior probabilities found for three classes for those learners.

	C_1	C_2	C_3
d_1	0.2	0.5	0.3
d_2	0.1	0.6	0.3
d_3	0.3	0.3	0.4

Calculate the posterior values for three classes for sum, median, minimum, maximum and product rules.

V. Question

Derive the update equations for K-class discrimination when the hidden units use tanh, instead of the sigmoid. Use the fact that $tanh' = (1 - tanh)^2$

VI. Question

Show the steps taken by the decision tree algorithm according to the following training examples.

Class	a_1	a_2
+	T	T
+	T	T
-	T	F
+	F	F
-	F	T
-	F	T

VII. Question

Derive the dual form L_d (Equation 13.19) of the nu-svm formulation (Equations 13.17, 13.18).